

# Rigorous Analysis of the Triangle Inequality for the NB Distance-Like Function

## Mathematical Proof and Disproof Report

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### Abstract

This report provides a complete, rigorous mathematical analysis of the triangle inequality for the proposed *Normalized Bounded (NB) distance-like function* defined in Equation (22) of the paper. We prove analytically that the triangle inequality holds strictly in the one-dimensional case ( $p = 1$ ). Conversely, we disprove the validity of the triangle inequality for all higher-dimensional spaces ( $p \geq 2$ ) by presenting a concrete, integer-based 2D counterexample. Finally, we provide the geometric intuition explaining why the triangle inequality is violated in higher dimensions.

## 1 Introduction and Definition

The  $p$ -dimensional NB dissimilarity function for two vectors  $x, y \in \mathbb{R}^p$  (not simultaneously zero) is defined as:

$$d(x, y) = \frac{1}{2} \frac{\sum_{i=1}^p |x_i - y_i| (|x_i| + |y_i|)}{\sum_{i=1}^p (x_i^2 + y_i^2)} \quad (1)$$

If  $x = y = 0$ , the distance is defined as  $d(0, 0) = 0$ .

## 2 Proof of the One-Dimensional Case ( $p = 1$ )

For  $p = 1$ , the function simplifies to:

$$d(x, y) = \frac{1}{2} \frac{|x - y| (|x| + |y|)}{x^2 + y^2} \quad (2)$$

Due to the scale-invariance of the function—namely  $d(kx, ky) = d(x, y)$  for all  $k \neq 0$ —and its negation symmetry  $d(-x, -y) = d(x, y)$ , we can assume without loss of generality that at least two of the three variables  $\{x, y, z\}$  are non-negative. This leads to two primary sign configurations to establish the triangle inequality:

$$d(x, z) \leq d(x, y) + d(y, z) \quad (3)$$

### 2.1 Piecewise Formulation by Relative Signs

For any  $u, v \in \mathbb{R}$ :

1. **Same Sign** ( $uv \geq 0$ ):

$$d(u, v) = \frac{1}{2} \frac{|u^2 - v^2|}{u^2 + v^2} \quad (4)$$

2. **Opposite Signs** ( $uv < 0$ ): Assuming  $u > 0$  and  $v < 0$ , let  $v = -b$  with  $b > 0$ . Then:

$$d(u, v) = \frac{1}{2} \frac{(u + b)^2}{u^2 + b^2} = \frac{1}{2} + \frac{|uv|}{u^2 + v^2} \quad (5)$$

## 2.2 Case A: $x, y, z > 0$ (All Positive)

Let  $A = x^2, B = y^2, C = z^2$ . Since  $x, y, z > 0$ , we have  $A, B, C > 0$ . The distances are:

$$d(x, y) = \frac{1}{2} \frac{|A - B|}{A + B}, \quad d(y, z) = \frac{1}{2} \frac{|B - C|}{B + C}, \quad d(x, z) = \frac{1}{2} \frac{|A - C|}{A + C} \quad (6)$$

We wish to prove:

$$\frac{|A - C|}{A + C} \leq \frac{|A - B|}{A + B} + \frac{|B - C|}{B + C} \quad (7)$$

By symmetry, assume without loss of generality that  $A \geq C$ . We analyze the three orderings of  $B$ :

### 2.2.1 Ordering 1: $A \geq B \geq C$

Here,  $|A - C| = A - C$ ,  $|A - B| = A - B$ , and  $|B - C| = B - C$ . The difference between the RHS and the LHS is:

$$\begin{aligned} \text{Diff} &= \left( \frac{A - B}{A + B} + \frac{B - C}{B + C} \right) - \frac{A - C}{A + C} \\ &= \frac{(A - B)(B + C) + (B - C)(A + B)}{(A + B)(B + C)} - \frac{A - C}{A + C} \\ &= \frac{2B(A - C)}{(A + B)(B + C)} - \frac{A - C}{A + C} \\ &= (A - C) \left[ \frac{2B(A + C) - (A + B)(B + C)}{(A + B)(B + C)(A + C)} \right] \end{aligned}$$

Expanding and factoring the numerator inside the brackets:

$$2AB + 2BC - (AB + AC + B^2 + BC) = AB + BC - AC - B^2 = (A - B)(B - C)$$

Thus, we obtain:

$$\text{Diff} = \frac{(A - C)(A - B)(B - C)}{(A + B)(B + C)(A + C)} \quad (8)$$

Since  $A \geq B \geq C > 0$ , all factors in the numerator are non-negative, and the denominator is strictly positive. Therefore,  $\text{Diff} \geq 0$ , and the inequality holds.

### 2.2.2 Ordering 2: $B \geq A \geq C$

Let  $f(u) = \frac{u - C}{u + C} = 1 - \frac{2C}{u + C}$  for  $u > 0$ . Since  $f'(u) = \frac{2C}{(u + C)^2} > 0$ ,  $f(u)$  is strictly increasing. Because  $B \geq A \geq C > 0$ , we have  $f(B) \geq f(A)$ , meaning:

$$\frac{A - C}{A + C} \leq \frac{B - C}{B + C} \leq \frac{B - A}{B + A} + \frac{B - C}{B + C} \quad (9)$$

which satisfies the inequality.

### 2.2.3 Ordering 3: $A \geq C \geq B$

Let  $g(u) = \frac{A - u}{A + u} = \frac{2A}{A + u} - 1$  for  $u > 0$ . Since  $g'(u) = -\frac{2A}{(A + u)^2} < 0$ ,  $g(u)$  is strictly decreasing. Because  $A \geq C \geq B > 0$ , we have  $g(C) \leq g(B)$ , meaning:

$$\frac{A - C}{A + C} \leq \frac{A - B}{A + B} \leq \frac{A - B}{A + B} + \frac{C - B}{C + B} \quad (10)$$

which satisfies the inequality.

### 2.3 Case B: $x, z > 0$ and $y < 0$ (Opposite Sign in the Middle)

Let  $A = x^2, B = y^2, C = z^2$ .

- The pairs  $(x, y)$  and  $(y, z)$  have opposite signs, so  $d(x, y) = \frac{1}{2} + \frac{|xy|}{A+B} > \frac{1}{2}$  and  $d(y, z) = \frac{1}{2} + \frac{|yz|}{B+C} > \frac{1}{2}$ .
- The pair  $(x, z)$  has the same sign, so  $d(x, z) = \frac{1}{2} \frac{|A-C|}{A+C} < \frac{1}{2}$  (since  $|A - C| < A + C$ ).

Therefore, we have:

$$d(x, z) < \frac{1}{2} < 1 < d(x, y) + d(y, z) \quad (11)$$

which holds with a strict margin of more than  $\frac{1}{2}$ .

### 2.4 Case C: $x, y > 0$ and $z < 0$ (Opposite Sign at One End)

Let  $z = -w$  with  $w > 0$ . We have  $x, y, w > 0$ . Let  $A = x^2, B = y^2, C = w^2$ . The distances are  $d(x, y) = \frac{1}{2} \frac{|A-B|}{A+B}$ ,  $d(y, z) = \frac{1}{2} + \frac{yw}{B+C}$ , and  $d(x, z) = \frac{1}{2} + \frac{xw}{A+C}$ . Subtracting  $\frac{1}{2}$  from both sides, the inequality simplifies to:

$$\frac{xw}{x^2 + w^2} \leq \frac{1}{2} \frac{|x^2 - y^2|}{x^2 + y^2} + \frac{yw}{y^2 + w^2} \quad (12)$$

By symmetry, assume  $x \geq y > 0$ . We want to show:

$$\frac{xw}{x^2 + w^2} - \frac{yw}{y^2 + w^2} \leq \frac{1}{2} \frac{x^2 - y^2}{x^2 + y^2} \quad (13)$$

We analyze two subcases based on  $w > 0$ :

#### 2.4.1 Subcase 1: $x \geq y \geq w > 0$

Let  $g(u) = \frac{uw}{u^2 + w^2}$  for  $u > 0$ . The derivative is  $g'(u) = w \frac{w^2 - u^2}{(u^2 + w^2)^2}$ . Since  $u \geq w$ , we have  $g'(u) \leq 0$ , meaning  $g(u)$  is strictly decreasing. Since  $x \geq y \geq w$ , we have  $g(x) \leq g(y)$ , which implies:

$$\frac{xw}{x^2 + w^2} - \frac{yw}{y^2 + w^2} \leq 0 \leq \frac{1}{2} \frac{x^2 - y^2}{x^2 + y^2} \quad (14)$$

which satisfies the inequality.

#### 2.4.2 Subcase 2: $w \geq x \geq y > 0$

By scale-invariance, let  $x' = x/y \geq 1$  and  $w' = w/y \geq x'$ . Let  $a = (x')^2 \geq 1$  and  $t = (w')^2 \geq a$ . The inequality is:

$$\frac{\sqrt{at}}{a+t} - \frac{\sqrt{t}}{1+t} \leq \frac{1}{2} \frac{a-1}{a+1} \quad (15)$$

Let  $\sqrt{t} = e^\theta$  and  $\sqrt{a} = e^\alpha$  where  $\theta \geq \alpha \geq 0$ . This can be rewritten using hyperbolic functions:

$$\frac{1}{2} \left[ \frac{1}{\cosh(\theta - \alpha)} - \frac{1}{\cosh \theta} \right] \leq \frac{1}{2} \frac{a-1}{a+1} \quad (16)$$

Since  $\cosh u$  is strictly increasing for  $u \geq 0$ , and for  $\theta \geq \alpha \geq 0$  we have  $\theta \geq \theta - \alpha \geq 0$ , we have:

- $\frac{1}{\cosh(\theta - \alpha)} \leq 1$
- $\cosh \theta \geq \cosh \alpha \implies -\frac{1}{\cosh \theta} \leq -\frac{1}{\cosh \alpha}$

Adding these bounds yields:

$$\frac{1}{\cosh(\theta - \alpha)} - \frac{1}{\cosh \theta} \leq 1 - \frac{1}{\cosh \alpha} = \frac{(\sqrt{a} - 1)^2}{a + 1} \quad (17)$$

We now verify if the upper bound satisfies the inequality:

$$\begin{aligned} \frac{1}{2} \frac{(\sqrt{a} - 1)^2}{a + 1} &\leq \frac{1}{2} \frac{a - 1}{a + 1} \iff (\sqrt{a} - 1)^2 \leq (\sqrt{a} - 1)(\sqrt{a} + 1) \\ &\iff \sqrt{a} - 1 \leq \sqrt{a} + 1 \\ &\iff -1 \leq 1 \end{aligned}$$

which is always true.

Thus, the triangle inequality holds rigorously in all cases for  $p = 1$ .

### 3 Disproof of the Multi-Dimensional Case ( $p \geq 2$ )

For dimensions  $p \geq 2$ , the triangle inequality is **not** preserved.

#### 3.1 2D Integer Counterexample ( $p = 2$ )

Let us define three vectors  $x, y, z \in \mathbb{R}^2$  as follows:

- $x = (-3, -1)$
- $y = (-3, 3)$
- $z = (1, 3)$

##### 3.1.1 Computation of $d(x, z)$ :

$$\begin{aligned} |x - z| &= (|-3 - 1|, |-1 - 3|) = (4, 4) \\ |x| + |z| &= (|-3| + |1|, |-1| + |3|) = (4, 4) \\ \text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |x_i - z_i|(|x_i| + |z_i|) = \frac{1}{2} [4(4) + 4(4)] = 16 \\ \text{Denominator} &= \sum_{i=1}^2 (x_i^2 + z_i^2) = [(-3)^2 + 1^2] + [(-1)^2 + 3^2] = 10 + 10 = 20 \\ d(x, z) &= \frac{16}{20} = 0.80 \end{aligned}$$

##### 3.1.2 Computation of $d(x, y)$ :

$$\begin{aligned} |x - y| &= (|-3 - (-3)|, |-1 - 3|) = (0, 4) \\ |x| + |y| &= (|-3| + |-3|, |-1| + |3|) = (6, 4) \\ \text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |x_i - y_i|(|x_i| + |y_i|) = \frac{1}{2} [0(6) + 4(4)] = 8 \\ \text{Denominator} &= \sum_{i=1}^2 (x_i^2 + y_i^2) = [(-3)^2 + (-3)^2] + [(-1)^2 + 3^2] = 18 + 10 = 28 \\ d(x, y) &= \frac{8}{28} = \frac{2}{7} \approx 0.2857 \end{aligned}$$

### 3.1.3 Computation of $d(y, z)$ :

$$\begin{aligned}
|y - z| &= (|-3 - 1|, |3 - 3|) = (4, 0) \\
|y| + |z| &= (|-3| + |1|, |3| + |3|) = (4, 6) \\
\text{Numerator} &= \frac{1}{2} \sum_{i=1}^2 |y_i - z_i|(|y_i| + |z_i|) = \frac{1}{2}[4(4) + 0(6)] = 8 \\
\text{Denominator} &= \sum_{i=1}^2 (y_i^2 + z_i^2) = [(-3)^2 + 1^2] + [3^2 + 3^2] = 10 + 18 = 28 \\
d(y, z) &= \frac{8}{28} = \frac{2}{7} \approx 0.2857
\end{aligned}$$

### 3.1.4 Comparison of Distances:

Sum of sub-distances:

$$d(x, y) + d(y, z) = \frac{2}{7} + \frac{2}{7} = \frac{4}{7} \approx 0.5714 \quad (18)$$

Direct distance:

$$d(x, z) = 0.80 \quad (19)$$

Comparing the two yields:

$$0.80 > 0.5714 \implies d(x, z) > d(x, y) + d(y, z) \quad (20)$$

Thus, the triangle inequality is violated by a margin of  $\approx 0.2286$ .

## 4 Geometric Intuition of the Violation

The failure in higher dimensions is caused by the combination of scale-normalization and axis-aligned path partitioning:

1. **Direct Path** ( $x \rightarrow z$ ): Both dimensions change simultaneously, meaning both components are active in the numerator while the total quadratic energy in the denominator is low (20).
2. **Partitioned Path** ( $x \rightarrow y \rightarrow z$ ): The step  $x \rightarrow y$  only changes component 2, while component 1 remains constant at  $-3$ . The step  $y \rightarrow z$  only changes component 1, while component 2 remains constant at 3.
3. In each sub-step, the constant (“idle”) component contributes heavily to the quadratic energy sum in the denominator ( $2 \times (-3)^2 = 18$  for the first step, and  $2 \times 3^2 = 18$  for the second step).
4. This “idle energy” inflates the denominators of the sub-distances, artificially shrinking  $d(x, y)$  and  $d(y, z)$  so that their sum falls below the direct distance.

## 5 Conclusion

While the NB distance-like function satisfies all other axioms of a metric space (non-negativity, identity of indiscernibles, and symmetry) and holds for the triangle inequality in 1D, it is **not** a valid metric in  $\mathbb{R}^p$  for  $p \geq 2$ . The authors of the paper only stress-tested the function on one-dimensional datasets and unfortunately generalized their conclusions to higher dimensions without checking vectors.